

MATH 221 Examination

Linear Algebra and Matrix Theory – Answer Key

1. (B) Its transpose equals its inverse.

An orthogonal matrix Q satisfies $Q^T Q = I$, which means $Q^T = Q^{-1}$. This property preserves lengths and angles, making orthogonal matrices represent rotations and reflections.

2. (B) The matrix A has full column rank.

When A has full column rank (rank n), the matrix $A^T A$ is invertible, ensuring the normal equations $A^T A x = A^T b$ have a unique solution. This is the least squares solution.

3. (B) All eigenvalues are real.

The Spectral Theorem guarantees that symmetric matrices have real eigenvalues and can be orthogonally diagonalized. However, eigenvalues need not be positive (they could be negative or zero) unless the matrix is positive definite.

4. (B) It decomposes A into a sequence of rotations, scaling, and another rotation.

The SVD $A = U \Sigma V^T$ can be interpreted as: first rotate by V^T , then scale by Σ , then rotate by U . This provides geometric insight into how A transforms space.

5. *Sample answer:*

This follows from the **Rank-Nullity Theorem**. For an $m \times n$ matrix A :

- $\text{rank}(A) + \text{nullity}(A) = n$
- The rank equals the dimension of the column space
- The nullity equals the dimension of the null space
- Therefore: $\dim(\text{Col}(A)) + \dim(\text{Null}(A)) = n$

This reflects that every column either contributes to the column space or corresponds to a free variable in the null space.

6. *Sample answer:*

The classical Gram-Schmidt process can suffer from:

- **Loss of orthogonality:** When vectors are nearly parallel, rounding errors accumulate, and computed vectors may not be truly orthogonal.
- **Numerical instability:** Later vectors in the sequence lose accuracy as errors compound.

Modified Gram-Schmidt addresses this by orthogonalizing each new vector against the already-computed orthogonal vectors sequentially, rather than against the original vectors. This reduces error propagation and maintains better numerical orthogonality.

7. *Sample answer:*

The eigenvalues of $A^T A$ are the squares of the singular values of A :

- If $A = U \Sigma V^T$, then $A^T A = V \Sigma^T U^T U \Sigma V^T = V \Sigma^2 V^T$
- Thus $A^T A$ has eigenvalues σ_i^2 where σ_i are singular values of A
- This shows that singular values are always non-negative

- The eigenvectors of $A^T A$ form the right singular vectors (columns of V)

This relationship is fundamental because it allows computation of SVD via eigenvalue decomposition.

8. *Sample answer:*

The condition number measures amplification of errors:

- A small change Δb in the right-hand side can cause a change Δx in the solution with $\|\Delta x\|/\|x\| \leq \kappa(A) \cdot \|\Delta b\|/\|b\|$
- When $\kappa(A)$ is large (ill-conditioned), small errors in data are magnified, leading to unreliable solutions
- For very large $\kappa(A)$, the matrix is nearly singular, and numerical solutions become unstable
- This is critical in applications like least squares regression and numerical PDEs